

18. Do the columns of  $B$  span  $\mathbb{R}^4$ ? Does the equation  $Bx=y$  have a solution for each  $y$  in  $\mathbb{R}^4$ ?

**Solution:** By Theorem 4, whether the columns of  $B$  span  $\mathbb{R}^4$  and whether the equation  $Bx=y$  have a solution for each  $y$  in  $\mathbb{R}^4$  is equivalent to checking whether each row of  $B$  has a pivot position. By defn. of pivot position, ~~each row~~ it corresponds to the leading 1 of a row of the matrix in reduced row echelon form.

$$\begin{array}{l}
 \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 1 & 2 & -3 & 7 \\ -2 & -8 & 2 & -1 \end{bmatrix} \xrightarrow[\text{+ Row 2}]{\text{Row 3} = \text{Row 3} - \text{Row 1}} \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 0 & -1 & -1 & 5 \\ -2 & -8 & 2 & -1 \end{bmatrix} \xrightarrow[\text{+ 2x Row 1}]{\text{Row 4} = \text{Row 4} + 2 \times \text{Row 1}} \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 0 & -1 & -1 & 5 \\ 0 & -2 & -2 & 3 \end{bmatrix} \\
 \xrightarrow[\text{+ Row 2}]{\text{Row 3} = \text{Row 3} + \text{Row 2}} \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & -2 & 3 \end{bmatrix} \xrightarrow[\text{+ 2x Row 1}]{\text{Row 4} = \text{Row 4} + 2 \times \text{Row 1}} \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -7 \end{bmatrix} \xrightarrow[\text{Interchange Row 3 and Row 4}]{\text{Row 2} = \text{Row 2} + 5 \times \text{Row 3}} \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
 \xrightarrow[\text{2x Row 3}]{\text{Row 1} = \text{Row 1} - 3 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{- 3x Row 2}]{\text{Row 1} = \text{Row 1} - 3 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{bmatrix}
 \end{array}$$

Verification of whether the final matrix is in reduced row echelon form

- Row 1, 2 and 3 are non-zero rows. Row 4 is a zero row.  $\therefore$  The condition that all non-zero rows must be above the zero rows is True.
  - Row 1 Leading Entry = 1 in Column 1  
Row 2 Leading Entry = 1 in Column 2  
Row 3 Leading Entry = 1 in Column 4
- $\therefore$  The condition that each row has its leading entry in a column to the right of the columns of the leading entries of all rows above it is True.